

UNIVERSITI TEKNOLOGI MALAYSIA

SECP3623 - DATABASE PROGRAMMING 2025/2026 - SEM 1

Lab Assignment 1 (5%) - Indexes

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B-TREE INDEXES

LAB 1

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PRESENTATION SLIDE

PRESENTATION TITLE / DESCRIPTION



B-TREE INDEX

A B-Tree index is a balanced tree structure that keeps data sorted for fast searching, inserting and deleting. It helps databases quickly locate rows without scanning the whole table, making queries more efficient

Internally, it has root, branch and leaf nodes. The leaf nodes store the actual data pointers, while the upper nodes guide the search path. All leaves are at the same level, keeping the tree balanced and ensuring fast access even as data grows.

Advantages and Disadvantages of B Tree Index

ADVANTAGES	DISADVANTAGES
Have a guaranteed time complexity of $O(\log n)$ for basic operation like insertion, deletion and searching	Based on disk-based data structures and can have a high disk usage
Self-balancing	Not best for all cases
High-concurrency and high-throughput	Might be slower compared to a binary search tree for small datasets

Result and Query time & Performance

```
CREATE INDEX idx_salary ON employee(salary);
SELECT * FROM employee WHERE salary BETWEEN 3000 AND 7000;
```

emp_id	emp_name	department	salary	join_date
1	Goh	HR	3000.00	2021-02-10
2	Dayang	Finance	7000.00	2022-03-11
3	Farra	Marketing	6000.00	2021-04-19
4	Safiya	IT	5000.00	2022-10-28
5	Syafiny	HR	3000.00	2021-02-10
6	Teh	HR	5000.00	2021-04-15
7	Lam	Sales	6000.00	2022-04-10
8	Yiya	Finance	6000.00	2022-04-10
9	Dania	Sales	3000.00	2023-05-09
10	Bella	Marketing	5000.00	2023-07-19
12	Firzana	HR	5000.00	2021-03-18
13	Yasmin	Sales	3000.00	2022-08-01
14	Boo	Marketing	4000.00	2021-11-11
15	Adriana	HR	3500.00	2021-11-11
16	Naim	Marketing	3000.00	2022-04-15
17	Mukhriz	HR	3000.00	2022-10-10
18	Alif	Finance	3000.00	2021-06-20
20	Afiq	Marketing	4000.00	2021-08-10

Before indexing:

0.01372225

After indexing:

0.00115875

```
CREATE INDEX idx_join_date ON employee(join_date);
SELECT * FROM employee WHERE join_date BETWEEN '2021-01-01' AND '2023-01-01';
```

emp_id	emp_name	department	salary	join_date
1	Goh	HR	3000.00	2021-02-10
2	Dayang	Finance	7000.00	2022-03-11
3	Farra	Marketing	6000.00	2021-04-19
4	Safiya	IT	5000.00	2022-10-28
5	Syafiny	HR	3000.00	2021-02-10
6	Teh	HR	5000.00	2021-04-15
7	Lam	Sales	6000.00	2022-04-10
8	Yiya	Finance	6000.00	2022-04-10
12	Firzana	HR	5000.00	2021-03-18
13	Yasmin	Sales	3000.00	2022-08-01
14	Boo	Marketing	4000.00	2021-11-11
15	Adriana	HR	3500.00	2021-11-11
16	Naim	Marketing	3000.00	2022-04-15
17	Mukhriz	HR	3000.00	2022-10-10
18	Alif	Finance	3000.00	2021-06-20
19	Iman	Sales	2000.00	2022-11-16
20	Afiq	Marketing	4000.00	2021-08-10

Before indexing:

0.00130900

After indexing:

0.00078125

Result and Query time & Performance

```
EXPLAIN SELECT * FROM employee WHERE salary BETWEEN 3000 AND 7000;
```

```
EXPLAIN SELECT * FROM employee WHERE join_date BETWEEN '2021-01-01' AND '2023-01-01';
```

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
▶	1	SIMPLE	employee	NULL	ALL	idx_join_date	NULL	NULL	NULL	20	85.00	Using where

```
DROP INDEX idx_salary ON employee;
```

```
DROP INDEX idx_join_date ON employee;
```

```
SET profiling = 1;
```

```
SELECT * FROM employee WHERE salary BETWEEN 3000 AND 7000;
```

```
SELECT * FROM employee WHERE join_date BETWEEN '2021-01-01' AND '2023-01-01';
```

```
SHOW PROFILES;
```

Query_ID	Duration	Query
1	0.00108100	SHOW WARNINGS
2	0.00031625	SET profiling = 1
3	0.00010300	SHOW WARNINGS
4	0.00077475	SELECT * FROM employee WHERE salary BETWEEN 3000 AND 7000 LIMIT 0, 1000
5	0.00041625	SELECT * FROM employee WHERE join_date BETWEEN '2021-01-01' AND '2023-01-01' LIMIT 0, 1000

Result and Query time & Performance

```
EXPLAIN SELECT * FROM employee WHERE salary BETWEEN 3000 AND 7000;
```

```
EXPLAIN SELECT * FROM employee WHERE join_date BETWEEN '2021-01-01' AND '2023-01-01';
```

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
▶	1	SIMPLE	employee	NULL	ALL	NULL	NULL	NULL	NULL	20	11.11	Using where

Why B-Tree Index is Ideal for Range and Sorting Queries ?

A B-Tree index keeps data in a sorted and balanced tree structure, which allows MySQL to quickly locate the start point of a range and then read the rest of the matching records in order. Because the data is already sorted inside the tree, B-Tree indexes are excellent for handling conditions like BETWEEN, >, <, or ORDER BY. This reduces query time and improves performance, especially for large datasets.

THANK YOU



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